Reverse the Queue

Category B Coding Assignment - Problem 3

Problem summary

Reverse a singly linked list in place. No new nodes may be created. The implementation handles empty and one-node lists.

Java solution

```
public class Main {
 static class Node {int data;Node next;Node(int d){data=d;}}
 static Node reverse(Node head){
  Node previous=null,current=head;
  while(current!=null){Node next=current.next;current.next=previous;previous=current;current=next;}
  return previous;
 }
 static Node build(int... a){Node h=null,t=null;for(int x:a){Node n=new Node(x);if(h==null)h=n;else t.next=n;t=n
 static void print(Node n){while(n!=null){System.out.print(n.data+" -> ");n=n.next;}System.out.println("null");}
 public static void main(String[] args){print(reverse(build(1,2,3,4)));print(reverse(build(9)));print(reverse(nu
}
```

Sample output

```
4 -> 3 -> 2 -> 1 -> null
9 -> null
null
```

Complexity

Time: O(n). Additional space: O(1). The previous, current and next references reverse each link.